DEVELOPER INFRASTRUCTURE PITCH DECK

rapid_ingestor

Kernel-Bypass, Zero-Allocation Telemetry & Log Ingestion Engine

Seastar C++ Reactor Framework

10x Vector Throughput



THE CORE INFRASTRUCTURE BOTTLENECK

Traditional Telemetry Middleware Hits the POSIX & Kernel Wall

Traditional Middleware (Vector / FluentBit)

- Kernel Context Switching: Millions of POSIX socket syscalls daily
- Thread Lock Contention: epoll and OS thread scheduling overhead
- Disk Flushing Stalls: OS page cache locks during burst writes
- Write Amplification: Direct unbatched raw logs choke ClickHouse

The rapid_ingestor Paradigm

- User-Space Networking: Zero kernel context switches on packet ingress
- Thread-Per-Core Locks: Isolated core execution without lock sharing
- Direct DMA Disk Access: Direct io_uring and O_DIRECT bypassing page cache
- Native Columnar Formatting: Zero fragmentation writes for ClickHouse

SEASTAR C++ CORE ARCHITECTURE

End-to-End Kernel Bypass & Zero-Allocation Pipeline

Kernel-Bypass Networking

DPDK and AF_XDP ring buffer ingress directly to user memory.

Thread-Per-Core Reactor

Share-nothing CPU pinning eliminating thread contention locks.

Direct Async Disk I/O

Linux io_uring and O_DIRECT bypassing OS page cache stalls.

EMPIRICAL PERFORMANCE PROOF

10x Throughput Advantage Over Vector

1.16M
Messages / sec steady ingestion throughput

>10x
Outperformance vs Vector rust runtime

237 MB/s
Deterministic flat-line wire bandwidth

Benchmarked on single core socket parsing with 204.7-byte average log payloads.

ANALYTICAL STORAGE OPTIMIZATION

Stream-to-Storage Processing Chain

STEP 01

Ring Ingress

Zero-copy kernel bypass packet capture.

STEP 02

Zero-Alloc Parse

Native protocol extraction in user memory.

STEP 03

Column Batch

Direct ClickHouse native block formatting.

STEP 04

Direct Commit

io_uring flush without page cache locks.

COMMERCIALIZATION STRATEGY

Open-Core Engine to Enterprise Control Plane

1. Open-Source Core

High-performance C++ ingestion engine distributed freely to build developer adoption.

Bottom-Up Dev Adoption

2. Enterprise Plugins

Monetize corporate identity extensions, LDAP input controls, and compliance tools.

Inbound Enterprise Pull

3. Managed Cloud

Hosted control plane managing multi-cluster ClickHouse and analytical pipelines.

SaaS Control Plane

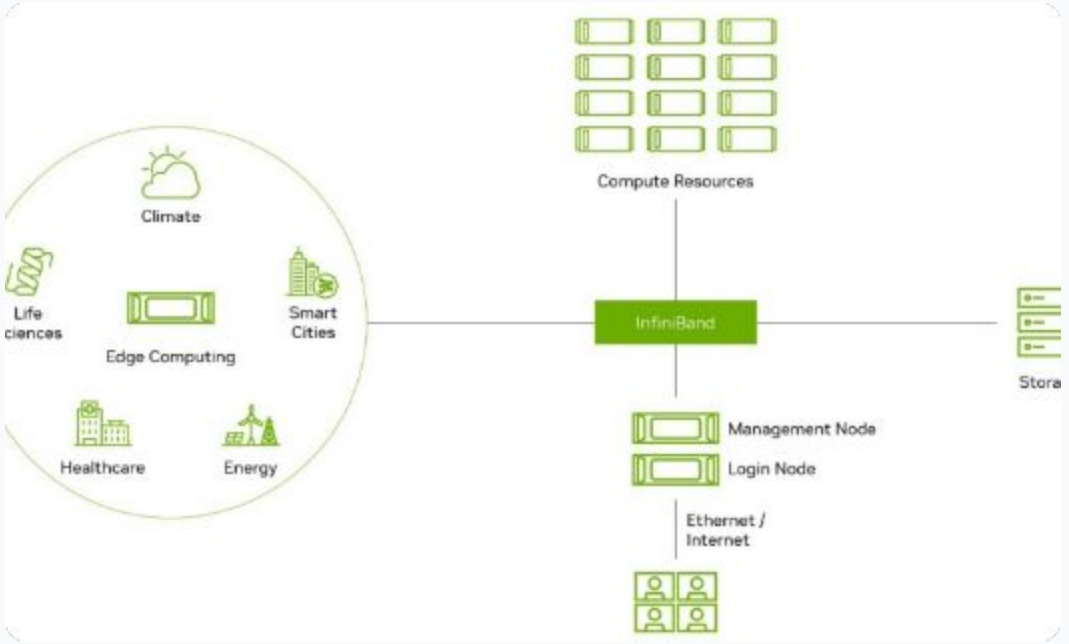
FOUNDER & TECHNICAL OWNERSHIP

3 Decades of Low-Level Systems Mastery

Solo Technical Founder

Built, optimized, and benchmarked rapid_ingestor independently.

- **Deep Systems Expertise:** Extensive track record in C++, Seastar, DPDK, AF_XDP, eBPF, and Go.
- **Datastore Mastery:** Direct experience optimizing ClickHouse and MongoDB storage boundaries.
- **Execution Power:** Demonstrated ability to single-handedly engineer breakthrough infrastructure.



Building the Next-Gen Telemetry Standard

Raising Day 0 / Pre-Seed capital to convert early inbound pull into enterprise design partnerships and scale core engineering.

MILESTONE 1

Design Partners

Deliver LDAP enterprise inputs for initial design cohorts.

MILESTONE 2

ClickHouse Ecosystem

Release native zero-copy ClickHouse block format plugins.

MILESTONE 3

Team Expansion

Recruit lead core systems engineer and GTM partner.

Image Sources



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